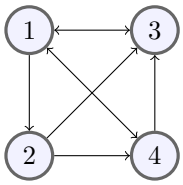
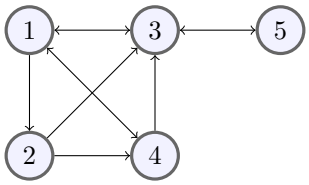
Homework 1- PageRank Exercises

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Exercise 1



(a) before



(b) after

$$\begin{bmatrix} 0 & 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 & 1 \\ 1/3 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \end{bmatrix}$$

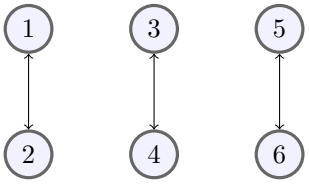
A matrix of (b)

It's easy to see that:

$$x_2 = \frac{x_1}{3},\ x_3 = \frac{x_1}{3} + \frac{x_2}{2} + \frac{x_4}{2} + x_5$$
$$x_4 = \frac{x_1}{3} + \frac{x_2}{2},\ x_5 = \frac{x_3}{2}$$

From which we can derive: $\frac{x_1}{49} \begin{bmatrix} 12 \\ 4 \\ 18 \\ 6 \\ 9 \end{bmatrix}$, hence showing: $rank(p_3) > rank(p_1)$

Exercise 2



(a)

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

(b)

$$x_1 = x_2,\ x_2 = x_1$$
$$x_3 = x_4,\ x_4 = x_3$$
$$x_5 = x_6,\ x_6 = x_5$$

(c)

The 3 subwebs form V_1 such that $dim(V_1) = 3$. This can easily be seen by looking at (c).

Exercise 3

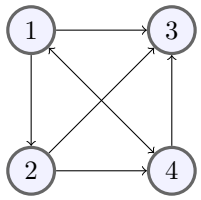
$V_1(A)$ is the subspace of all the eigenvectors of A associated to eigenvalue 1. Its dimension is equivalent to the geometric multiplicity of the eigenvalue $\lambda = 1$, i.e. the number of linearly independent distributions representing a possible ranking configuration.

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1/3 \\ 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/3 \\ 0 & 0 & 1/2 & 0 & 1/3 \\ 1/2 & 0 & 1/2 & 1/2 & 0 \end{bmatrix}$$

$$x_1 = x_2 + \frac{x_5}{3},\ x_2 = \frac{x_1}{2}$$
$$x_3 = \frac{x_4}{2} + \frac{x_5}{3},\ x_4 = \frac{x_3}{2} + \frac{x_5}{3}$$
$$x_5 = \frac{x_1}{2} + \frac{x_3}{2} + \frac{x_4}{2}$$

$$V_1(A) = x_5 \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \\ 2/3 \\ 1 \end{bmatrix}$$

Exercise 4



(a) Web graph structure

$$A = \begin{bmatrix} 0 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

(b) Link matrix A

Given a generic strongly connected web, we remove a link from a page to another one, not satisfying anymore the necessary condition for strong connectivity. To solve this problem we identify the Perron eigenvalue $\lambda = 0.5614$ and its corresponding eigenvector.

$$A - \lambda I = \begin{bmatrix} -\lambda & 0 & 0 & 1/2 \\ 1/3 & -\lambda & 0 & 0 \\ 1/3 & 1/2 & -\lambda & 1/2 \\ 1/3 & 1/2 & 0 & -\lambda \end{bmatrix} \qquad (A - \lambda I)x = \begin{bmatrix} -\lambda x_1 + \frac{x_4}{2} \\ \frac{x_1}{3} - \lambda x_2 \\ \frac{x_1}{3} + \frac{x_2}{2} - \lambda x_3 + \frac{x_4}{2} \\ \frac{x_1}{3} + \frac{x_2}{2} - \lambda x_4 \end{bmatrix} \qquad x_4 \begin{bmatrix} 0.207 \\ 0.123 \\ 0.439 \\ 0.232 \end{bmatrix}$$

Normalized rankings with respect to $\lambda = 0.5614$

The ranking is reasonable as page 3, that has the highest number of incoming links, has the highest score. Page 4 is the second most important page as there are 2 links pointing to it; however, it also spreads its importance score to page 3 so its final importance score is penalized. Likewise, even though page 1 and page 2 have the same number of incoming links, page 1's score is slightly penalized by its higher number of outgoing links.

Exercise 5

The importance score of a page is computed by means of the i-th element of the eigenvector corresponding to the eigenvalue $\lambda = 1$. To prove that in any web the importance score of a page with no back-links is zero, we execute the generic procedure for finding this particular eigenvector x and observe its behavior.

Given a generic web of pages, one of which has no back-links, it can be represented by the matrix A, with a null row j in correspondence of the page that receives no links

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}$$

We compute the importance score by solving the linear system $(A - I)x = 0$

$$\begin{bmatrix} a_{11} - 1 & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} - 1 & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} - 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_j \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Since we are interested in the value of the importance score of the page with no links, we only compute x_j, given by the formula

$$x_j = \sum_{i=1}^{n} a_{ji} x_i$$

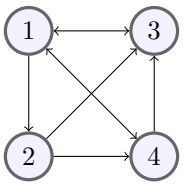
Given that all the elements of the j-th row, with the exception of the j-th column, are equal to zero, we have just one contribution to the sum:

$$-1 \cdot x_j = 0 \Rightarrow x_j = 0$$

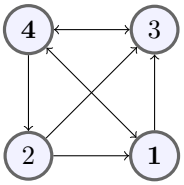
Exercise 6

(a)

Given a generic web W represented by figure a , we swap pages **1** and **4** and obtain the graph represented in figure b



(a) Original graph



(b) Modified graph

$$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & 1 & 0 \end{bmatrix}$$

Just by observing the two matrices it's possible to observe that $\tilde{\text{A}}$ is similar to the original matrix A, with the exception for some rows and columns that seem to be swapped. To show these changes in a more rigorous way we introduce the permutation matrix $P_{1,4}$ obtained by swapping the 1st and 4th row (or column) of the identity matrix I.

We then proceed by executing the product $\hat{A} = PAP$ in two steps

$$P_{1,4} \cdot A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{2} & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$$

By left-multiplying the matrix A with the permutation matrix we obtain an intermediate result in which only rows 1 and 4 are swapped (all links pointing to page 4 are now pointing to page 1 and vice versa). This operation corresponds to swapping the sources of the links. We then proceed by right-multiplying this intermediate result by the same permutation matrix $P_{1,4}$

$$\hat{A} = P_{1,4}A \cdot P_{1,4} = \begin{bmatrix} \frac{1}{3} & \frac{1}{2} & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & 1 & 0 \end{bmatrix}$$

This second product swaps the columns, which translates into swapping the destination nodes of the links.

(b)

Now that we have shown that after swapping two nodes i and j of a graph we obtain a representation matrix equivalent to a permutation of the previous one, we are going to prove that, if x is an eigenvector of A, then the vector y obtained by multiplying the permutation matrix P and x, is indeed an eigenvector of the permuted matrix $\tilde{\text{A}}$ with eigenvalue $\lambda = 1$.

We know that x is an eigenvector of A iff

$$Ax = \lambda x$$

We left-multiply both sides of the equations by a generic permutation matrix P

$$PAx = \lambda Px$$

We replace $y = Px$ and obtain

$$PAx = \lambda y$$

By definition we know that $y = Px$, so we compute $x = P^{-1}y$. Since P is a permutation matrix it's also orthogonal, from which we deduce that $P^{-1} = P^T$. If, like in this case, we are swapping just two rows at a time, P is also symmetric, so $P^T = P$. We proceed by replacing $x = P^{-1}y = P^T y = Py$

$$PAPy = \lambda y$$

Since $\tilde{A} \coloneqq PAP$ with proper substitutions we have

$$\tilde{A}y = \lambda y \quad \square$$

(c)

In point (b) we demonstrated that $\mathbf{Px}$ is an eigenvector of $\tilde{\mathbf{A}}$, and one can immediately see that the only effect that $\mathbf{P}$ has on the ranking is a change of indices whereas the relative ranking remains untouched. Furthermore, All permutations of the identity matrix are always invertible, so the similarity relationship between $\mathbf{A}$ and $\tilde{\mathbf{A}}$ can always be maintained. Finally, it should be noted that similar matrices have the same eigen-multiplicities, so $\tilde{\mathbf{A}}$ retains the unique ranking property of $\mathbf{A}$.

Exercise 7

Given a matrix $A \in R^{n \times n}$ representing the pages composing a web with no dangling nodes, we can state that A is *column stochastic*.

If we consider another matrix $S \in R^{n \times n}$ such that

$$s_{ij} \coloneqq \frac{1}{n}$$

it's easy to show that S is *column stochastic* as well

$$\sum_{i=1}^{n} s_{ij} = \sum_{i=1}^{n} \frac{1}{n} = n \cdot \frac{1}{n} = 1 \qquad \forall j = 1, \dots, n$$

We then proceed by constructing a third matrix M, defined as a convex combination of A and S as follows

$$M = (1 - m)A + mS \qquad m \in \mathbb{R} : 0 \leq m \leq 1$$

In order to prove that the matrix $M \in \mathbb{R}^{n \times n}$ is *column stochastic*, we have to show that all its entries are non-negative **(a)** and that the sum of its elements over columns is equal to 1 **(b)**.

(a) Non-negativity: A is constructed in such a way that

$$a_{ij} \geq 0 \qquad \forall\, i, j = 1, \dots, n$$

The elements of S are positive by definition:

$$s_{ij} \coloneqq \frac{1}{n} \geq 0 \qquad \forall\, i, j = 1, \dots, n$$

Both m and $(1 - m)$ are non-negative real numbers:

$$m \in [0, 1] \quad \Rightarrow \quad (1 - m) \in [0, 1]$$

Each element of the matrix M is defined as

$$m_{ij} = (1 - m)a_{ij} + ms_{ij}$$

Since we are summing two non-negative numbers, each defined as a product of two non-negative numbers, we have

$$m_{ij} \geq 0 \qquad \forall\, i, j = 1, \dots, n \qquad \square$$

(b) Column sums: As we stated before, each element of the matrix is defined as

$$m_{ij} = (1 - m)a_{ij} + ms_{ij}$$

We now compute the sum of all the entries belonging to the same column j

$$\sum_{i=1}^{n} m_{ij} \quad = \quad \sum_{i=1}^{n}(1-m)a_{ij} + \sum_{i=1}^{n} m s_{ij} \quad =$$

Since m and $1-m$ are two constants, they can be taken out of the sum operator

$$= \quad (1-m)\sum_{i=1}^{n} a_{ij} + m\sum_{i=1}^{n} s_{ij}$$

By hypothesis both A and S are column stochastic, implying that

$$\sum_{i=1}^{n} s_{ij} = 1 \qquad \text{and} \qquad \sum_{i=1}^{n} a_{ij} = 1 \qquad \forall j = 1, \ldots, n$$

From which we derive

$$\sum_{i=1}^{n} m_{ij} \quad = (1-m)\cdot 1 + m \cdot 1 = 1 - m + m = 1 \qquad \forall\, i, j = 1, \ldots, n \qquad \square$$

Exercise 8

Given any two compatible column-stochastic matrices, we can show that:

$$\begin{bmatrix} a_{11} & \cdots & \cdots & a_{1n} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & \cdots & b_{1n} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ b_{n1} & \cdots & \cdots & b_{nn} \end{bmatrix} = b_{11}(\, a_{11} + \ldots + a_{n1}\,) + \ldots + b_{n1}(\, a_{1n} + \ldots + a_{nn}\,)$$

sum of two different columns of **A**

Which then reduces down to 1.

sum of the elements in the first column of matrix **B**

$$\sum_{i=1} AB_{i1} = b_{11}(1) + \ldots + b_{n1}(1) = \sum_{i=1} b_{i1} \; = 1$$

This generalizes to the sum of every single column of AB; therefore, proving that it in fact is column-stochastic.

Exercise 9

Let:

$$\mathbf{Z} = (1-m)\mathbf{A}$$

Using:

$$\mathbf{Z}\mathbf{x} + m\mathbf{s} = \mathbf{x} \tag{1}$$

and the last row of our $\mathbf{Z}$ matrix, which comes as a consequence of a page with no backlinks:

$$\begin{bmatrix} 0 & \ldots & 0 \end{bmatrix}$$

We can then rearrange (1) to get the following:

$$\mathbf{Z}\mathbf{x} - \mathbf{x} = -m\mathbf{s}$$

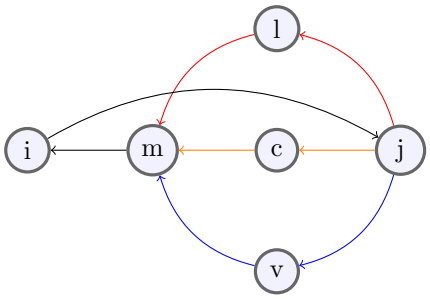
$$\mathbf{x}(\mathbf{Z} - \mathbf{I}) = -m\mathbf{s}$$

The LHS of which will give us a product for which we only care about the product between the last row of $\mathbf{Z} - \mathbf{I}$ and $\mathbf{x}$, namely:

$$-x_n = -\frac{m}{n} \;\longleftrightarrow\; x_n = \frac{m}{n}$$

Exercise 10

(a-b)



pictorial representation of a possible web

By induction on k. For $k = 1$, we have that any path a_{ij}, where $i \neq j$, will necessarily have a positive value iff there is a link between nodes i and j. This is almost the definition of the adjacency matrix of a directed graph, and is the basis to our proof. Assuming that a_{ij}^k is some positive number. We show that a_{ij}^{k+1} is also some non-zero positive number. In fact since $A^{k+1} = A^k \cdot A$, then

$$a_{ij}^{k+1} = a_{i1}^{(k)} a_{1j} + a_{i2}^{(k)} a_{2j} + ... + a_{in}^{(k)} a_{nj} = \sum_{m=1}^{n} a_{im}^k a_{mj}$$

It can be seen that if there is a link from j to m, then $(A^{k+1})_{ij}$ will also be a positive number. This sum will necessarily be null if there are no links from j to m, and then from m to i. More simply, if the result of the summation above is 0, then there exists no $k+1$ length path from j to i.

(c)

We know that $(A^k)_{ij}$ will be a positive number if there are a series of links that lead to i from j. Naturally, the sum $\sum_{i=0}^{p} A^i$ will be greater than 0 iff we can get to i from j using anywhere from 0 to p number of links.

(d)

In a strongly connected web of pages, any one page is reachable from some other page. This implies that there is always a path to a page, from some other.

$$\mathbf{W} = (\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + ... + \mathbf{A}^{n-1})$$

One should note that, a path of length n would necessarily imply a revisit of some node. Any tour covering all unique nodes of the graph will at most have a length of $n - 1$. In other words, including terms equal to A^n or greater would be entirely redundant. Therefore, W will be a positive matrix iff we have at least a strongly connected web of pages.

(e)

In exercise 8, we reached the conclusion that the product of two column-stochastic matrices is column-stochastic. The identity matrix itself is also automatically column-stochastic as each unique column contains just one value with value 1. We use matrix $\mathbf{W}$ from **(d)**, and we show the sum of all elements in the n_{th} column of $\frac{1}{n}\mathbf{W}$ as

$$\frac{1}{n}\sum_{i=1}^{n} W_{in} = \frac{1}{n}(\sum_{i=1}^{n} I_{in} + \sum_{i=1}^{n} A_{in} + \sum_{i=1}^{n} A_{in}^2 + ... + \sum_{i=1}^{n} A_{in}^{n-1}) = \frac{1}{n}(1_1 + 1_2 + 1_3 + ... + 1_n) = 1$$

This is because all the matrices used in the sum are column-stochastic themselves.

(f)

First, we show that if $x \in V_1(\mathbf{A})$, then $x \in V_1(\mathbf{B})$.

$$\mathbf{Bx} = \mathbf{x}$$

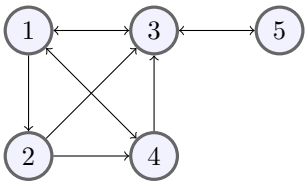
$$\frac{\mathbf{1}}{n}(\mathbf{Ix} + \mathbf{Ax} + \mathbf{A^2x} + ... + \mathbf{A^{n-1}x}) = \mathbf{x}$$

We know that 1 is an eigenvalue of all the matrices within the parenthesis. So, we can replace the LHS terms with 1$\mathbf{x}$ to get:

$$\frac{\mathbf{1}}{n}(\sum_{i}^{n} \mathbf{x}) = \frac{\mathbf{1}}{n} n\mathbf{x} = \mathbf{x}$$

We have proven that $\mathbf{B}$ or $\frac{1}{n}\mathbf{W}$ is positive and column stochastic. $\mathbf{B}$ has $dim(V_1(B)) = 1$ by Lemma 3.2, and this has the immediate implication that $dim(V_1(\mathbf{A})) = 1$. This is because if $\mathbf{x}$ consisted of two or more independent vectors, then it would mean that $dim(V_1(\mathbf{B}))$ would be greater than 1. However, we know that $dim(V_1(\mathbf{B})) = 1$, and this implies that $dim(V_1(\mathbf{A})) = 1$.

Exercise 11



(a) modified 2.1

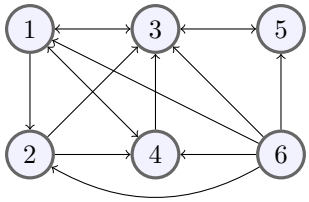
$$\begin{bmatrix} 0.03 & 0.03 & 0.455 & 0.455 & 0.03 \\ 0.3133 & 0.03 & 0.03 & 0.03 & 0.03 \\ 0.3133 & 0.455 & 0.03 & 0.455 & 0.88 \\ 0.3133 & 0.455 & 0.03 & 0.03 & 0.03 \\ 0.03 & 0.03 & 0.455 & 0.03 & 0.03 \end{bmatrix}$$

(b) $\mathbf{M}$ matrix

$$x_u \begin{bmatrix} 0.237 \\ 0.097 \\ 0.349 \\ 0.138 \\ 0.178 \end{bmatrix}$$

The results were obtained by running matrix $\mathbf{A}$ through a function in Python.

Exercise 12



(a) modified 2.1

$$\begin{bmatrix} 0 & 0 & 1/2 & 1/2 & 0 & 1/5 \\ 1/3 & 0 & 0 & 0 & 0 & 1/5 \\ 1/3 & 1/2 & 0 & 1/2 & 1 & 1/5 \\ 1/3 & 1/2 & 0 & 0 & 0 & 1/5 \\ 0 & 0 & 1/2 & 0 & 0 & 1/5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(b) $\mathbf{A}$ matrix

$$\begin{bmatrix} 1/40 & 1/40 & 9/20 & 9/20 & 1/40 & 39/200 \\ 0.3083 & 1/40 & 1/40 & 1/40 & 1/40 & 39/200 \\ 0.3083 & 9/20 & 1/40 & 9/20 & 7/8 & 39/200 \\ 0.3083 & 9/20 & 1/40 & 1/40 & 1/40 & 39/200 \\ 1/40 & 1/40 & 9/20 & 1/40 & 1/40 & 39/200 \\ 1/40 & 1/40 & 1/40 & 1/40 & 1/40 & 1/40 \end{bmatrix}$$

(c) $\mathbf{M}$ matrix

With the $\mathbf{A}$ matrix, we're able to get the following ranking:

$$\frac{1}{2.717}\begin{bmatrix} 0.667 & 0.22 & 1 & 0.33 & 0.5 & 0 \end{bmatrix}^T$$

In comparison to the ranking that we're able to get with $\mathbf{M}$:

$$\begin{bmatrix} 0.231 & 0.095 & 0.340 & 0.135 & 0.174 & 0.025 \end{bmatrix}^T$$

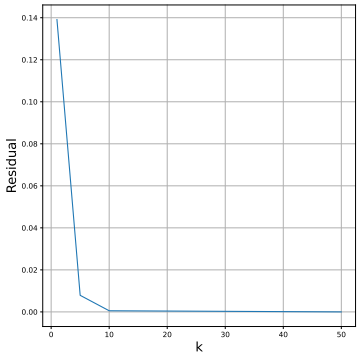
The ranking orders remain; however, the ranking that we're able to achieve with just $\mathbf{A}$ gives zero importance to page 6; whereas, the $\mathbf{M}$ ranking gives some weight also to page 6.

Exercise 14

Taking into account the matrix defined in exercise 11, we want to compute its eigenvectors observing how the values of $||M^k x_0 n - q||_1$ and $\frac{||M^k x_0 - q||_1}{||M^{k-1} x_0 - q||_1}$ change with iterations. Given the initial guess of x_0 chosen randomly we get the following results $x_0 = \begin{bmatrix} 0.774 & 0.439 & 0.859 & 0.697 & 0.094 \end{bmatrix}^T$

k	$\|M^k x_0 - q\|_1$	$\frac{\|M^k x_0 - q\|_1}{\|M^{k-1} x_0 - q\|_1}$
1	1.39e-1	nan
5	7.89e-3	4.31e-1
10	5.43e-4	5.77e-1
50	1.26e-13	5.72e-1

(a) Residuals and ratios at different iterations



(b) Convergence rate

We observe that the residual decreases a lot and converges to zero, meaning that after enough iteration we get a good approximation of the exact eigenvector q. From theoretical results we know that the ratio between the second and first largest eigenvalues $\frac{|\lambda_2|}{|\lambda_1|}$ dictates the rate of convergence. Since $|\lambda_1| = 1$, the ratio $\frac{||M^k x_0 - q||_1}{||M^{k-1} x_0 - q||_1}$ should converge to $|\lambda_2|$. We can observe that, as expected, the ratio converges towards the second largest eigenvalue of q: 0.574.

The value of $c = \max|1 - 2\min_{1 \leq i \leq n} M_{ij}|$ was found to be $c = 0.94$.

Exercise 16

Given a generic matrix $A \in \mathbb{R}^{n \times n}$, A is not diagonalizable iff the geometric multiplicity of at least one of its eigenvalues is less than its algebraic multiplicity.

- Algebraic multiplicity of λ_i is the integer α such that $p(\lambda) = (\lambda - \lambda_i)^\alpha q(\lambda)$ with $q(\lambda) \neq 0$
- Geometric multiplicity of λ_i is the number of linearly independent eigenvectors corresponding to λ_i

Since this is a necessary condition to guarantee diagonalizability of any matrix, if we want to show that $M = (1 - m)A + mS$ is non diagonalizable it's sufficient to find one eigenvalue of M whose algebraic and geometric multiplicity don't coincide.

$$M = (1-m)\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \\ 1 & \frac{1}{2} & 0 \end{bmatrix} + m\begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{m}{3} & \frac{3-m}{6} & \frac{3-m}{6} \\ \frac{m}{3} & \frac{m}{3} & \frac{3-m}{6} \\ 1 - \frac{2m}{3} & \frac{3-m}{6} & \frac{m}{3} \end{bmatrix}$$

The symbolic eigenvalues and respective eigenvectors of M are found to be

- $\lambda_1 = 1$ with algebraic multiplicity 1
  - $v_1 = [\frac{m-3}{2m-4}, \frac{-1}{m-2}, 1]^T$
- $\lambda_2 = (\frac{m}{2} - \frac{1}{2})$ with algebraic multiplicity 2
  - $v_2 = [0, -1, 0]^T$

  Both eigenvectors have geometric multiplicity equal to one. Showing that the second eigenvalue has different algebraic and geometric multiplicities proves that, for any value of $m \in [0, 1)$ the matrix M is not diagonalizable.

Exercise 17

The damping factor m is introduced to solve the problem regarding disconnected graphs. At each step the surfer goes towards one of the pages connected to the one he is currently on. If we are dealing with a disconnected graph at some point we might get stuck. To solve this issue we modify A into M, thus having a $1-m$ probability of following an existing link, and a m probability of visiting a random page.

- $m = 0$: No stochasticity is added to the choice of the next page to go to
- $m = 1$: A's initial structure will have no contribution at all in the definition of the ranking, so at each step the surfer will be redirected to the next step randomly generating the same score fo all pages
- Intermediate values of m will yield intermediate solutions. Lower values of m will give more importance to the web's structure, while higher values will promote a more equal treatment to all pages